

$U \leq k^n$: k -vector subspace

SI

k^m

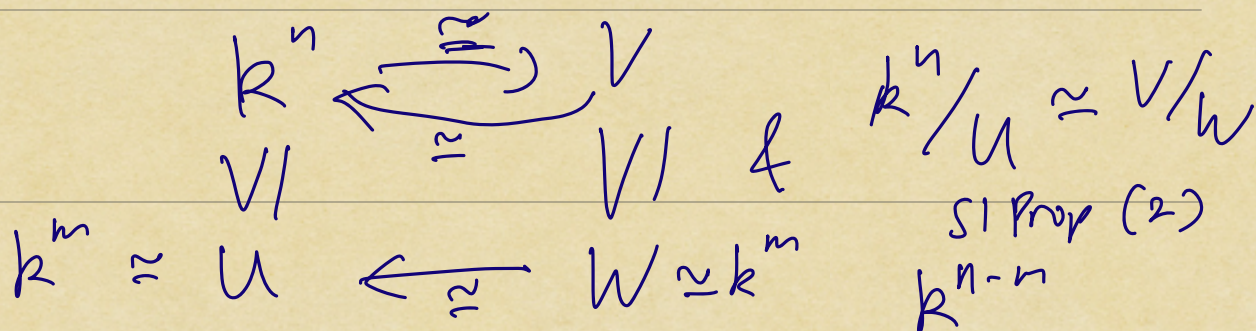
Proposition (2) $k^n/U \cong k^{n-m}$

Consequences:

(1) V : n -dim'l k -vector space
 V/U
 W : m -dim'l k -subspace

Then $\dim_k(V/U) = n - m$

Pf:



$k^n/U \cong V/U$
 SI Prop (2)
 k^{n-m}

(2) (Rank-nullity)

$$T: V \longrightarrow W$$

then $\dim_k \ker T + \dim_k \operatorname{im} T = \dim_k V$

Follows from (1) using

factoring triangle

(i.e. "first isom. thm.")

Solving linear systems of equations

V, W : finite dim'l k -vector spaces

$$T: V \longrightarrow W$$

Q: For $w \in W$, solve for

$$v \text{ in } T(v) = w.$$

Method:

$$\begin{array}{ccc} K^m & \xrightarrow[u_1]{\cong} & V \\ \downarrow & & \downarrow T \\ \underbrace{u_2^{-1} \circ T \circ u_1}_{\uparrow} & & K^n \xrightarrow[u_2]{\cong} W \end{array}$$

represented
by an $n \times m$
matrix $A \in M_{n \times m}(K)$

Solving $T(v) = w$

$$B^{-1} \cdot \left(\begin{array}{l} \Leftarrow \\ \Leftarrow \end{array} \right) \boxed{\begin{array}{l} Ax = u_2^{-1}(w) = y \\ BAx = By \end{array}}$$

For this, we can multiply
on the left by any invertible
matrix $B \in M_{n \times n}(K)$

Equivalently, we can modify
the system up to row operations

Example:

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 1 & 1 & -1 & 1 \\ -1 & 1 & 3 & 0 \end{bmatrix}$$

$$y = \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix}$$

Solve

$$Ax = y \quad \leadsto \quad [A | y]$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & 3 \\ 1 & 1 & -1 & 1 & -2 \\ -1 & 1 & 3 & 0 & 0 \end{array} \right]$$

$$\downarrow \begin{array}{l} R_2 \mapsto R_2 - R_1 \\ R_3 \mapsto R_3 + R_1 \end{array}$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & 3 \\ 0 & -1 & -1 & 1 & -5 \\ 0 & 3 & 3 & 0 & 3 \end{array} \right]$$

$$\downarrow R_1 \mapsto R_1 + 2R_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -2 & 2 & -7 \\ 0 & -1 & -1 & 1 & -5 \\ 0 & 3 & 3 & 0 & 3 \end{array} \right]$$

$$\begin{aligned} & \downarrow R_3 \mapsto R_3 + 3R_2. \\ & \downarrow R_2 \mapsto (-1) \times R_2 \end{aligned}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -2 & 2 & -7 \\ 0 & 1 & 1 & -1 & 5 \\ 0 & 0 & 0 & 3 & -12 \end{array} \right]$$

$$\text{char} \neq 3 \quad \swarrow \quad R_3 \mapsto 3^{-1} R_3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -2 & 2 & -7 \\ 0 & 1 & 1 & -1 & 5 \\ 0 & 0 & 0 & 1 & -4 \end{array} \right] \quad \dim_n W$$

$$\searrow \quad \text{char} = 3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & -1 & -7 \\ 0 & 1 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

REF free

$$R_2 \mapsto R_2 + R_3$$

$$R_1 \mapsto R_1 - 2 \cdot R_3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -2 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -4 \end{array} \right] \text{ RREF}$$

$\downarrow$ free

$$x_1 - 2x_3 = 1$$

$$x_2 + x_3 = 1$$

$$x_4 = -4$$

Solutions:

$$x_1 = 1 + 2s$$

$$x_2 = 1 - s$$

$$x_3 = s \in k \text{ arbitrary}$$

$$x_4 = -4$$

$$\dim_k(\text{im } T) = 3$$

T is surjective

$$\begin{aligned} x_1 + x_3 - x_4 &= -1 \\ x_2 + x_3 - x_4 &= -1 \end{aligned}$$

Solutions:

$$x_1 = -1 - s + t$$

$$x_2 = -1 - s + t$$

$$\left. \begin{aligned} x_3 &= s \\ x_4 &= t \end{aligned} \right\} \text{ arbitrary}$$

$$\dim_k(\text{im } T) = 2 < 3$$

T is not surjective

Caution:

$$\underbrace{A}_{(n \times m)} \underbrace{x}_{k^m} = \underbrace{y}_{k^n} \quad (\neq)$$

$(n \times m) \times (m \times 1) = n \times 1$

Column operations

$$A C x = C y$$

for $C \in GL_m(k)$

doesn't make sense unless $n=m$

$$\prod C^{-1}$$

$$C^{-1} A C x = y$$

Observation

$$T(v) = w$$

$$\Downarrow$$

$$A x = y$$

$$A \in M_{n \times m}(k)$$

Then: Once we reduce A to reduced row echelon form (RREF); i.e. once, using row operations, we get an

Equation of the form

$$A'x' = y'$$

Where A' has the following features:

(1) In each row of A' with a non-zero entry the first entry is 1. This is a leading 1.

(2) If a column contains a leading 1, then all other entries in that column are 0s.

(3) Leading 1s go from

top left to bottom right
as one goes down the
rows of the matrix

then the solutions are
obtained as follows:

(i) identify inconsistencies
i.e. rows of 0s in A'
matched up with non-zero
entries in y'

(ii) identify columns without
leading 1s (free variables)

(iii) Solve for remaining
variables in terms
of free variables

Observation (2)

When $w=0$, solving

$$T(v) = 0$$

$(\Leftrightarrow)$ Describing $\ker T$

$$(\Leftrightarrow) Ax = 0$$

$$(\Leftrightarrow) A'x' = 0$$

In this case, we find

$$\dim_n \ker T = \# \text{ of free variables}$$

And Rank-Nullity says

$$\dim_n \operatorname{im} T = \# \text{ of leading 1s}$$